



Department of Computer & Software Engineering

CE150L: Linear Circuit Analysis Lab

Lab Project

3 Level Automatic Light Control Circuit

Submitted by:

Syed Zulkifal

BSCE 23053

Zain Rashid

BSCE 23041

Haseeb Ashfaq

BSCE 23048

Introduction:

The 3 Level Automatic Light Control Circuit project aims to develop a smart light control system that adjusts the number of lights based on the availability of light. This circuit offers energy efficiency by automatically turning on the appropriate number of lights as it gets darker and turning off all lights when light is fully available.

Technical Specifications:

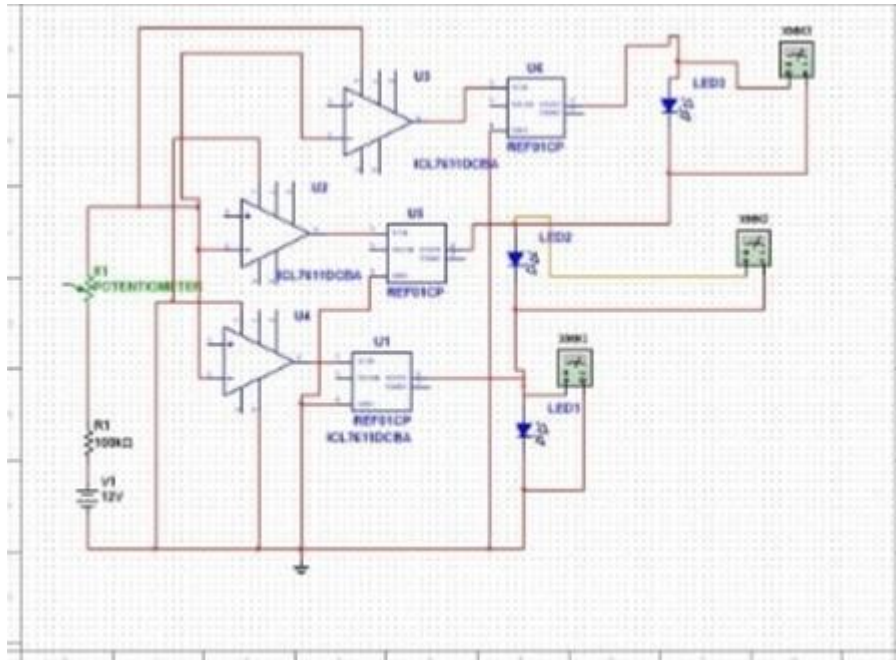
- The 3 Level Automatic Light Control Circuit operates within the following technical specifications:
- Power Supply: +12V
- Light Sensing Range: Daylight to Darkness
- Operating Temperature: 0°C to 50°C
- Active Power Consumption: Varies based on the number of lights turned on

Components Used:

- The circuit utilizes the following components:
- Light-dependent resistor (LDR)
- Diodes
- 220 ohm resistors
- 10k Ohm resistors
- 1k resistors
- Potentiometer 10k
- relay
- capacitor 100 uF
- operational amplifier
- Resistors (1K, 10K, etc.)
- LEDs (3)
- Power supply (+12V DC)

- Other Basic Components (Wires, connectors, etc.)

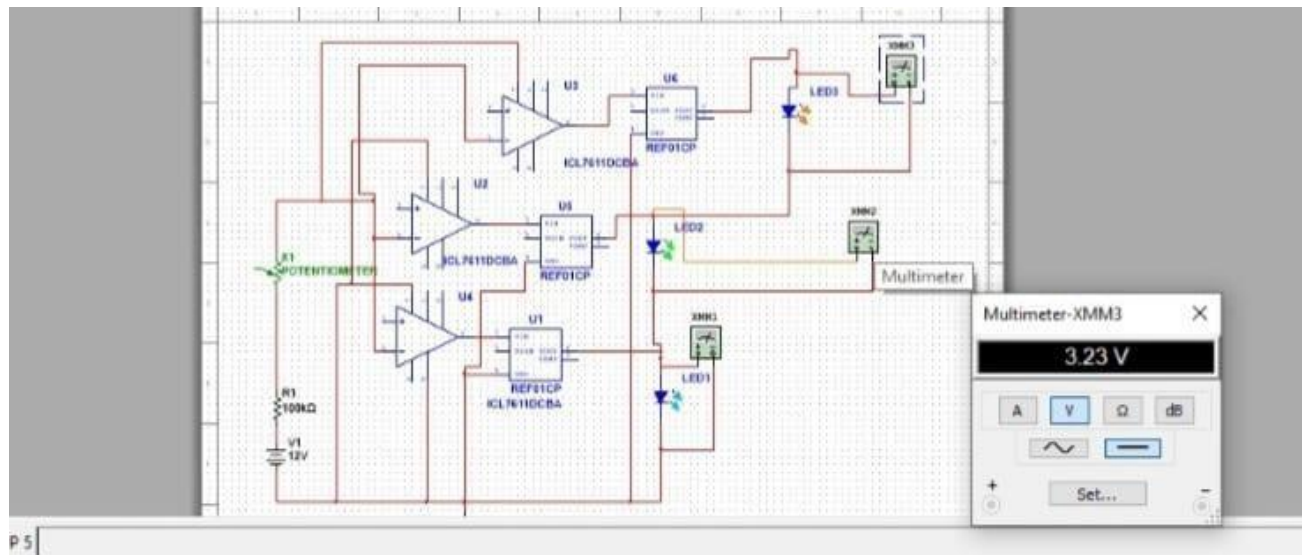
Circuit Design & Working:



The 3 Level Automatic Light Control Circuit employs a voltage divider circuit, transistors, and resistors to achieve intelligent light control. The circuit works as follows:

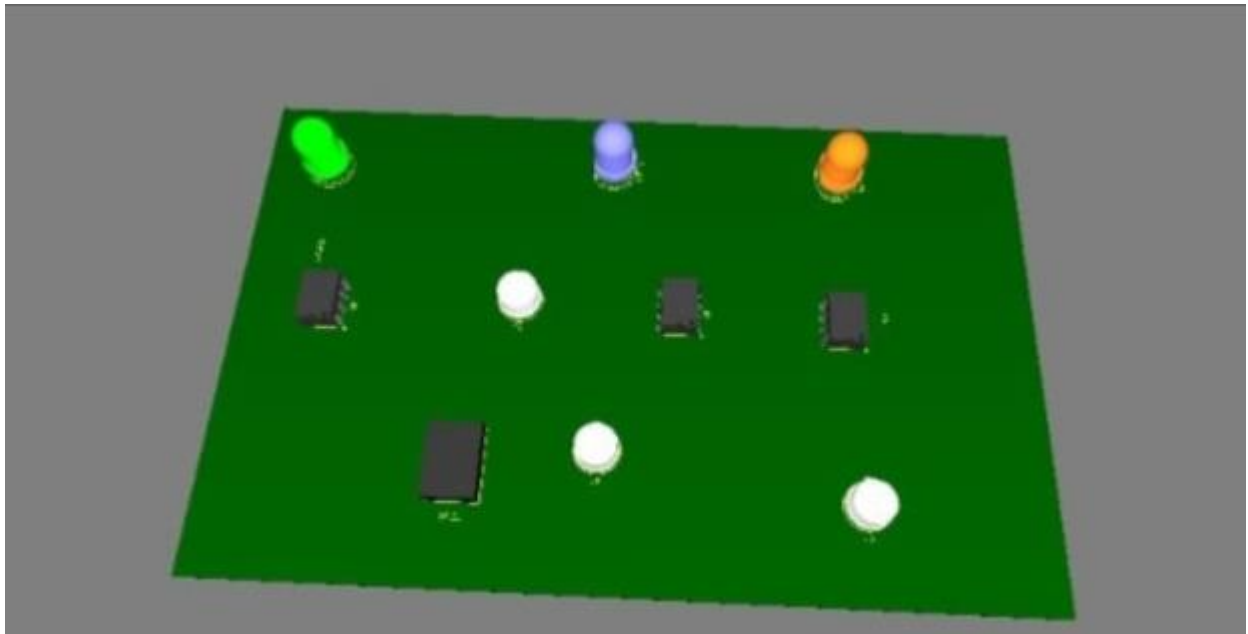
- The LDR senses the intensity of light.
- As the light intensity decreases, the resistance of the LDR increases.
- The voltage across the LDR is compared to a reference voltage using a voltage divider circuit.
- Based on the comparison, the appropriate number of transistors are activated to turn on the corresponding number of LEDs.
- When the light intensity increases, the resistance of the LDR decreases, and the transistors turn off the LEDs accordingly.

Simulation Results:



Simulation tests were conducted to verify the functionality of the 3 Level Automatic Light Control Circuit. Screenshots of simulations for various light intensity scenarios demonstrate the expected behavior of the circuit.

Hardware Implementation:



The hardware implementation of the 3 Level Automatic Light Control Circuit involves translating the designed schematic into a physical circuit on a veroboard or PCB. Key aspects of the hardware implementation include veroboard/PCB layout, component placement, wiring connections, power supply integration, and testing.

Results & Observations

Upon successful implementation and testing, the 3 Level Automatic Light Control Circuit demonstrated efficient light control based on ambient light conditions. The circuit accurately adjusted the number of lights according to the availability of sunlight, providing energy savings and convenience.

Conclusion

The 3 Level Automatic Light Control Circuit offers an intelligent solution for lighting systems by automatically adjusting the number of lights based on ambient light conditions. This project successfully demonstrates how technology can be used to optimize energy usage and enhance user convenience in various applications, such as outdoor lighting systems, garden lights, and streetlights.